LQR with Optimal Feedback Control (OFC)

Mode: `lqr-applied-to-both-cart-manip`

2D Welded Cart-Pendulum System with Muscle Dynamics and Joint-Space Manipulator Control

Research Note

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Contents

1 Overview

This note describes the formulation and implementation of the `lqr-applied-to-both-cart-manip` simulation mode. The key architectural difference from the `lqr-applied-to-cart-manip-following-cart` mode is that the **cart body is rigidly welded to the manipulator end-effector (EE)**. There are therefore no prismatic actuators on the cart: the cart position is *entirely determined* by the manipulator's joint angles via forward kinematics.

1. **Manipulator (2 DOF):** A 2-link revolute arm whose EE is driven by an OFC chain (LQR → Muscle → ZFT → IK → Computed-Torque) to reach the target (x^*, y^*).
2. **Cart-Pendulum (2 DOF):** A 3D gimbal pendulum welded to the manipulator EE. It receives no direct actuation — the pendulum is stabilised indirectly through the motion of the EE (and hence the cart body).

The two sub-systems share a single Drake `MultibodyPlant` with *separate model instances* but a rigid weld constraint connecting the cart body to the EE frame.

Key contrast with the other mode:

Aspect	**cart-manip-following-cart**	**both-cart-manip**
Cart actuation	Prismatic (impedance force)	None (welded to EE)
Manipulator role	Passive tracker via IK + CTC	Active driver of cart
Impedance block	✓ present	× removed
Cart $[x, y]$	Plant state (prismatic DOF)	FK output: $\mathrm{FK}(q_1, q_2)$
LQR state space	14D (cart + pend + muscle + ZFT)	14D (arm + pend + muscle + ZFT)

2 State and Coordinate Definitions

2.1 Plant State (8D)

Because the cart is welded to the EE, the plant no longer contains prismatic DOF for the cart. Instead the plant state is:

$$\boldsymbol{s}_{\text{plant}} = \underbrace{[q_1,\ q_2]^\top}_{\text{arm joints}} \oplus \underbrace{[\alpha,\ \beta]^\top}_{\text{pend. angles}} \oplus \underbrace{[\dot{q}_1,\ \dot{q}_2]^\top}_{\text{arm vel.}} \oplus \underbrace{[\dot{\alpha},\ \dot{\beta}]^\top}_{\text{pend. ang. vel.}} \in \mathbb{R}^8$$

where q_1 is the base-to-link1 joint angle and q_2 is the link1-to-link2 joint angle. Note that **Drake returns joints in URDF parse order**, which for this URDF is $[q_2, q_1]$; index conversion is handled internally.

2.2 EE / Cart Position (computed)

The cart body position in the world frame is *not* a state variable; it is computed from the arm state via forward kinematics:

$$\boldsymbol{p} = [x, y]^\top = f_{\text{FK}}(q_1, q_2) \in \mathbb{R}^2$$
$$\dot{\boldsymbol{p}} = J(q_1, q_2)\, [\dot{q}_1, \dot{q}_2]^\top, \quad J = \frac{\partial f_{\text{FK}}}{\partial \boldsymbol{q}} \in \mathbb{R}^{2\times 2}$$

A dedicated Drake `LeafSystem` (`EndEffectorKinematics2D`) evaluates these at every integration step and outputs $(\boldsymbol{p}, \dot{\boldsymbol{p}})$.

2.3 Full OFC / Augmented State (14D)

The LQR operates on the augmented state:

$$\boldsymbol{x} = \left[q_1,\ q_2,\ \alpha,\ \beta,\ \dot{q}_1,\ \dot{q}_2,\ \dot{\alpha},\ \dot{\beta},\ F_x,\ F_y,\ x_{\text{ref}},\ y_{\text{ref}},\ \dot{x}_{\text{ref}},\ \dot{y}_{\text{ref}}\right]^\top \in \mathbb{R}^{14}$$

where the first 8 entries come from the plant, the next 2 from the muscle dynamics, and the last 4 from the ZFT reference mass.

2.4 Manipulator State (4D)

$$\boldsymbol{s}_{\text{manip}} = [q_1,\ q_2,\ \dot{q}_1,\ \dot{q}_2]^\top \in \mathbb{R}^4$$

3 Control Architecture

3.1 Signal Chain

All actuation flows through the manipulator joint torques:

$$\textbf{LQR} \xrightarrow{\boldsymbol{u}} \textbf{Muscle} \xrightarrow{\boldsymbol{F}} \textbf{ZFT} \xrightarrow{\boldsymbol{s}_{\text{ref}}} \textbf{IK} \xrightarrow{\boldsymbol{q}_d,\dot{\boldsymbol{q}}_d,\ddot{\boldsymbol{q}}_d} \textbf{CTC} \xrightarrow{\boldsymbol{\tau}} \textbf{Manipulator Plant} \xrightarrow{\boldsymbol{p}=f_{\text{FK}}} \text{(cart moves)}$$

The pendulum angles α, β are shaped *indirectly* by the cart (EE) acceleration that results from the joint torques — there is no direct torque input to the pendulum.

3.2 Block Descriptions

Block 1 — Muscle Dynamics

$$\boxed{\dot{F}_i = \frac{-F_i + u_i}{\tau_m}, \quad i \in \{x, y\}}$$

- **Input:** $\boldsymbol{u} \in \mathbb{R}^2$ (LQR neural command)
- **Output:** $\boldsymbol{F} = [F_x, F_y]^\top \in \mathbb{R}^2$ (muscle force)
- **Time constant:** $\tau_m = 0.03$ s

Block 2 — ZFT Reference Mass

The muscle force drives a virtual reference mass that provides the smooth EE target trajectory. Its spring-damper coupling is to the *EE position* $\boldsymbol{p} = f_{\text{FK}}(\boldsymbol{q})$, not to the plant cart state (there is no prismatic cart state).

$$\boxed{M_{\text{ref}}\,\ddot{\boldsymbol{p}}_{\text{ref}} = \boldsymbol{F} + K\,(\boldsymbol{p} - \boldsymbol{p}_{\text{ref}}) + D\,(\dot{\boldsymbol{p}} - \dot{\boldsymbol{p}}_{\text{ref}})}$$

- **Inputs:** $\boldsymbol{F}$ (muscle); $\boldsymbol{p}, \dot{\boldsymbol{p}}$ (EE kinematics via FK)
- **Output:** $\boldsymbol{s}_{\text{ref}} = [x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]^\top \in \mathbb{R}^4$; also $\ddot{\boldsymbol{p}}_{\text{ref}} \in \mathbb{R}^2$ for feed-forward
- **Parameters:** $M_{\text{ref}} = 1.0$ kg, $K = 50$ N/m, $D = 10$ N $\cdot$ s/m

Block 3 — IK Solver (ZFT Joint Reference)

Rather than a one-shot IK per step, a *differential IK* integrates joint velocity commands resulting from the ZFT output:

$$\boxed{\dot{\boldsymbol{q}}_d = J^\dagger(\boldsymbol{q})\left[\,K_p(\boldsymbol{p}_{\text{ref}} - \boldsymbol{p}) + \dot{\boldsymbol{p}}_{\text{ref}}\,\right] + J^\dagger(\boldsymbol{q})\,\ddot{\boldsymbol{p}}_{\text{ref}}\,\Delta t}$$

where $J^\dagger$ is the Moore–Penrose pseudoinverse of the 2×2 Jacobian. The joint position reference $\boldsymbol{q}_d$ is obtained by integration. This approach avoids the overhead of a full nonlinear IK solve at every step.

- **Inputs:** $\boldsymbol{p}_{\text{ref}}, \dot{\boldsymbol{p}}_{\text{ref}}, \ddot{\boldsymbol{p}}_{\text{ref}}$ (ZFT outputs); plant state (for J and warm-start)
- **Outputs:** $\boldsymbol{q}_d, \dot{\boldsymbol{q}}_d, \ddot{\boldsymbol{q}}_d \in \mathbb{R}^2$

Block 4 — Finite-Horizon LQR Controller

The LQR minimises the quadratic cost over a finite horizon T:

$$\boxed{J = \int_0^T \left[\boldsymbol{e}^\top Q\,\boldsymbol{e} + \boldsymbol{u}^\top R\,\boldsymbol{u}\right] dt + \boldsymbol{e}(T)^\top Q_N\,\boldsymbol{e}(T)}$$

where $\boldsymbol{e} = \boldsymbol{x} - \boldsymbol{x}^*$ is the 14D state error from the goal:

$$\boldsymbol{x}^* = [q_1^*, q_2^*, 0, 0, 0, 0, 0, 0, 0, 0, x^*, y^*, 0, 0]^\top$$

with (q_1^*, q_2^*) solved analytically from the target (x^*, y^*) via 2-link IK. The backward Riccati ODE gives the time-varying gain:

$$\boldsymbol{u}(t) = -R^{-1}B^\top P(t)\,\boldsymbol{e}(t)$$

Cost matrices:

$$\begin{aligned} Q &= \operatorname{diag}(100, 100, 500, 500, 50, 50, 100, 100, 0.1, 0.1, 1, 1, 1, 1) \\ Q_N &= 2\,Q \\ R &= \operatorname{diag}(10, 10) \end{aligned}$$

The larger R (compared to the other mode) penalises neural effort more heavily, giving smoother joint motion.

Block 5 — Joint-Space Computed-Torque Controller

$$\boxed{\boldsymbol{\tau} = M(\boldsymbol{q})\,\ddot{\boldsymbol{q}}_d + C(\boldsymbol{q}, \dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} + G(\boldsymbol{q}) + K_p(\boldsymbol{q}_d - \boldsymbol{q}) + K_d(\dot{\boldsymbol{q}}_d - \dot{\boldsymbol{q}})}$$

- **Inputs:** $[\boldsymbol{q}_d, \dot{\boldsymbol{q}}_d, \ddot{\boldsymbol{q}}_d]$ (IK); $[\boldsymbol{q}, \dot{\boldsymbol{q}}]$ (plant)
- **Output:** $\boldsymbol{\tau} \in \mathbb{R}^2$ (joint torques, capped at ± 100 N $\cdot$ m)
- **Gains:** $K_p = 200$ N $\cdot$ m/rad, $K_d = 60$ N $\cdot$ m $\cdot$ s/rad

The torques $\boldsymbol{\tau}$ are the **only** actuation input to the plant.

4 Block Diagram

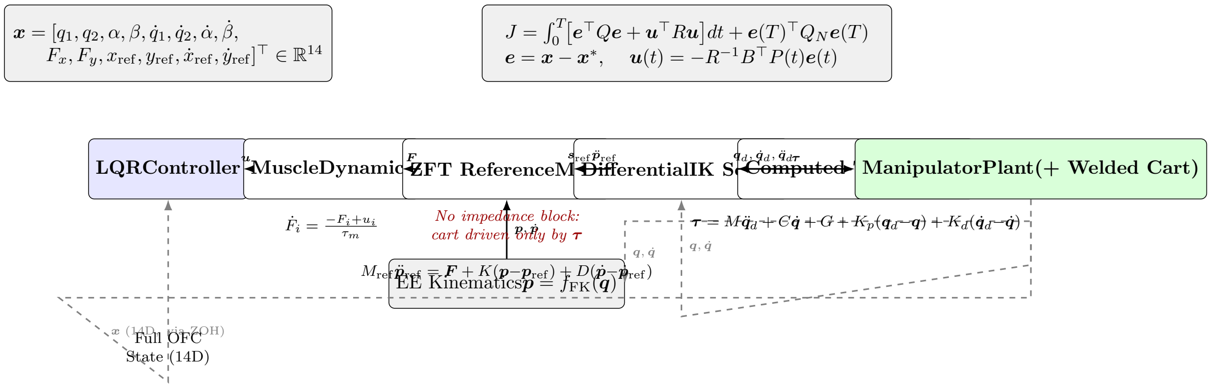

Figure 1: Block diagram of the `lqr-applied-to-both-cart-manip` mode. The cart body is welded to the manipulator EE, so there are no prismatic actuators. All actuation is through joint torques $\boldsymbol{\tau}$. Dashed arrows: feedback and warm-start signals. The EE kinematics block converts arm joint angles to task-space position $\boldsymbol{p} = f_{\mathrm{FK}}(\boldsymbol{q})$, which replaces the cart state in the ZFT ODE.

5 Signal Flow Summary

Table 1: Signal connections in the control loop (`lqr-applied-to-both-cart-manip`).

From	To	Signal	Dim.
LQR Controller	Muscle Dynamics	$\boldsymbol{u}$ (neural command)	$\mathbb{R}^2$
Muscle Dynamics	ZFT Reference Mass	$\boldsymbol{F}$ (muscle force)	$\mathbb{R}^2$
Muscle Dynamics	State Mux (14D)	$\boldsymbol{F}$ (for LQR state)	$\mathbb{R}^2$
EE Kinematics	ZFT Reference Mass	$\boldsymbol{p}, \dot{\boldsymbol{p}}$ (EE as cart)	$\mathbb{R}^4$
ZFT Reference Mass	IK Solver	$\boldsymbol{p}_{\mathrm{ref}}, \dot{\boldsymbol{p}}_{\mathrm{ref}}, \ddot{\boldsymbol{p}}_{\mathrm{ref}}$	$\mathbb{R}^6$
ZFT Reference Mass	State Mux (14D)	$\boldsymbol{s}_{\mathrm{ref}}$	$\mathbb{R}^4$
IK Solver	Computed-Torque Ctrl	$\boldsymbol{q}_d, \dot{\boldsymbol{q}}_d, \ddot{\boldsymbol{q}}_d$	$\mathbb{R}^6$
Manipulator Plant	Computed-Torque Ctrl	$[\boldsymbol{q}, \dot{\boldsymbol{q}}]$ (arm state)	$\mathbb{R}^4$
Computed-Torque Ctrl	Manipulator Plant	$\boldsymbol{\tau}$ (joint torques)	$\mathbb{R}^2$
Manipulator Plant	EE Kinematics	$\boldsymbol{q}, \dot{\boldsymbol{q}}$ (plant state)	$\mathbb{R}^4$
Manipulator Plant	IK Solver	plant state (warm-start)	$\mathbb{R}^8$
Manipulator Plant	State Mux	$\boldsymbol{s}_{\mathrm{plant}}$ (8D)	$\mathbb{R}^8$
State Mux	ZOH (10 ms)	$\boldsymbol{x}$ (14D)	$\mathbb{R}^{14}$
ZOH	LQR Controller	$\hat{\boldsymbol{x}}$ (held)	$\mathbb{R}^{14}$

6 Linearised System

The LQR is designed on the 14D augmented linearisation obtained from the welded plant. The equilibrium $(\boldsymbol{x}^*, \boldsymbol{u}^* = \boldsymbol{0})$ is:

$$\boldsymbol{x}^* = [q_1^*, q_2^*, 0, 0, 0, 0, 0, 0, 0, 0, x^*, y^*, 0, 0]^\top$$

where (q_1^*, q_2^*) is solved analytically from the 2-link inverse kinematics for the target (x^*, y^*). Around this equilibrium:

$$\dot{\boldsymbol{x}} = A\,\boldsymbol{x} + B\,\boldsymbol{u}$$

The matrices $A \in \mathbb{R}^{14\times 14}$ and $B \in \mathbb{R}^{14\times 2}$ are obtained by `build_linearized_for_complete_system_2d()`, which uses Drake's automatic Jacobian (via `Linearize`) on the welded plant augmented with muscle and ZFT dynamics.

The Riccati equation is solved backward in time from $t = T$ to $t = 0$ at $\Delta t = 10$ ms to obtain the gain schedule $K(t) = R^{-1}B^\top P(t)$, which is interpolated online.

7 Simulation Parameters

Table 2: Default simulation parameters.

Parameter	**Value**	**Description**
Arm link 1 length	≈ 2.20 m	From URDF geometry
Arm link 2 length	≈ 1.25 m	From URDF geometry
Pendulum mass	0.3 kg	
Pendulum length	0.25 m	
Arm damping	$0.1\ \mathrm{N\cdot s/rad}$	Both joints
Pendulum damping	$0.1\ \mathrm{N\cdot s/rad}$	Both gimbal axes
τ_m	0.03 s	Muscle time constant
M_{ref}	1.0 kg	ZFT reference mass
K	50 N/m	ZFT impedance stiffness
D	$10\ \mathrm{N\cdot s/m}$	ZFT impedance damping
K_p	$200\ \mathrm{N\cdot m/rad}$	CTC position gain
K_d	$60\ \mathrm{N\cdot m\cdot s/rad}$	CTC velocity gain
$\tau_{\max}$	$100\ \mathrm{N\cdot m}$	Joint torque limit
LQR horizon T	10.0 s	
Δt (LQR)	10 ms	Riccati integration step
Simulation Δt	1 ms	Drake integrator step

8 Implementation Notes

- **Weld constraint**: The cart body is added to the plant inside `SystemBuilder.build()` when `weld_cart_to_manip_ee=True`. Drake enforces the constraint algebraically so the cart position is always equal to the EE pose with zero constraint force residual.
- **No impedance block**: Unlike the companion mode, the impedance force block (`ImpedanceForce2D`) is **not instantiated**. The sole actuation path is joint torques $\boldsymbol{\tau}$.

- **Differential IK**: The `ZFTJointReferenceIK` system uses the velocity-integration method (`ik_method="differential"`) with position feedback gain $K_p = 10\ \mathrm{s}^{-1}$ rather than solving a full nonlinear IK problem at every step. This avoids kinematic singularities and is faster.
- **EE kinematics**: The `EndEffectorKinematics2D` system computes $(\boldsymbol{p}, \dot{\boldsymbol{p}})$ from the plant context at every step using Drake's `CalcPointsPositions` and the geometric Jacobian.
- **Drake joint ordering**: The URDF returns joints in parse order $[q_2, q_1]$. All user-facing code uses $[q_1, q_2]$ order; conversions happen inside `CupManipulator`.
- **ZOH**: A Zero-Order Hold at 10 ms on the 14D augmented state breaks the algebraic loop between the LQR and the continuous plant/muscle/ZFT dynamics.

9 Equation Flow Summary

This section traces the complete signal chain from sensor reading to actuator output at every simulation step.

Step-by-step signal chain

(1) Read plant state. The plant outputs its 8D state:

$$\boldsymbol{s}_{\text{plant}}(t) = [\, q_1,\ q_2,\ \alpha,\ \beta,\ \dot{q}_1,\ \dot{q}_2,\ \dot{\alpha},\ \dot{\beta}\,]^\top \in \mathbb{R}^8$$

(2) EE kinematics — compute cart position via FK. The arm joint angles $\boldsymbol{q} = [q_1, q_2]^\top$ are passed to the EE kinematics block:

$$\boldsymbol{p}(t) = f_{\text{FK}}(\boldsymbol{q}), \qquad \dot{\boldsymbol{p}}(t) = J(\boldsymbol{q})\,\dot{\boldsymbol{q}}$$

This $(\boldsymbol{p}, \dot{\boldsymbol{p}})$ substitutes for the cart state in all downstream blocks — the welded cart has no independent position state.

(3) Muscle dynamics — low-pass filter on neural command.

$$\dot{\boldsymbol{F}}(t) = \frac{-\boldsymbol{F}(t) + \boldsymbol{u}(t)}{\tau_m} \quad \Longrightarrow \quad \boldsymbol{F}(t) \in \mathbb{R}^2$$

$\boldsymbol{F}$ is fed to the ZFT reference mass and also stacked into the 14D state.

(4) ZFT reference mass — smooth end-effector target.

$$\ddot{\boldsymbol{p}}_{\text{ref}}(t) = \frac{1}{M_{\text{ref}}}\big[\boldsymbol{F}(t) + K\big(\boldsymbol{p}(t) - \boldsymbol{p}_{\text{ref}}(t)\big) + D\big(\dot{\boldsymbol{p}}(t) - \dot{\boldsymbol{p}}_{\text{ref}}(t)\big)\big]$$

Integration gives:

$$\boldsymbol{s}_{\text{ref}}(t) = [\, x_{\text{ref}},\ y_{\text{ref}},\ \dot{x}_{\text{ref}},\ \dot{y}_{\text{ref}}\,]^\top \in \mathbb{R}^4$$

All three outputs $(\boldsymbol{p}_{\text{ref}}, \dot{\boldsymbol{p}}_{\text{ref}}, \ddot{\boldsymbol{p}}_{\text{ref}})$ feed the IK solver.

(5) Full 14D state assembly.

$$\boldsymbol{x}(t) = \underbrace{[q_1, q_2, \alpha, \beta]}_{\text{joint/pend pos.}} \oplus \underbrace{[\dot{q}_1, \dot{q}_2, \dot{\alpha}, \dot{\beta}]}_{\text{joint/pend vel.}} \oplus \underbrace{[F_x, F_y]}_{\text{muscle}} \oplus \underbrace{[x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]}_{\text{ZFT ref.}}$$

A ZOH ($\Delta = 10$ ms) samples this to $\hat{\boldsymbol{x}}$.

(6) Finite-horizon LQR — optimal neural command.

$$\boldsymbol{u}(t) = -\underbrace{R^{-1}B^\top P(t)}_{K(t)}\big(\hat{\boldsymbol{x}}(t) - \boldsymbol{x}^*\big) \in \mathbb{R}^2$$

This closes the outer loop back to Step 3 (muscle input) at the next step.

(7) Differential IK — ZFT target to joint references.

$$\dot{\boldsymbol{q}}_d(t) = J^\dagger(\boldsymbol{q})\big[K_p(\boldsymbol{p}_{\text{ref}} - \boldsymbol{p}) + \dot{\boldsymbol{p}}_{\text{ref}}\big] + J^\dagger(\boldsymbol{q})\,\ddot{\boldsymbol{p}}_{\text{ref}}\,\Delta t$$

$$\boldsymbol{q}_d(t) = \boldsymbol{q}_d(t{-}\Delta t) + \dot{\boldsymbol{q}}_d(t)\,\Delta t$$

(8) Computed-torque controller — joint torques.

$$\boldsymbol{\tau}(t) = M(\boldsymbol{q})\ddot{\boldsymbol{q}}_d + C(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + G(\boldsymbol{q}) + K_p(\boldsymbol{q}_d - \boldsymbol{q}) + K_d(\dot{\boldsymbol{q}}_d - \dot{\boldsymbol{q}}) \in \mathbb{R}^2$$

$\boldsymbol{\tau}$ is the **sole actuation signal** applied to the welded plant. The joint motion moves the EE (and the welded cart) toward the target, while the pendulum is stabilised indirectly through the resulting EE acceleration.

Compact causal chain

$$\boldsymbol{s}_{\text{plant}} \xrightarrow{\text{FK}} \boldsymbol{p}, \dot{\boldsymbol{p}} \xrightarrow[\text{ZFT}]{\boldsymbol{F}} \boldsymbol{p}_{\text{ref}}, \dot{\boldsymbol{p}}_{\text{ref}}, \ddot{\boldsymbol{p}}_{\text{ref}} \xrightarrow{\text{diff. IK}} \boldsymbol{q}_d, \dot{\boldsymbol{q}}_d, \ddot{\boldsymbol{q}}_d \xrightarrow{\text{CTC}} \boldsymbol{\tau} \longrightarrow \text{plant}$$

$$\boldsymbol{s}_{\text{plant}} \oplus \boldsymbol{F} \oplus \boldsymbol{s}_{\text{ref}} \xrightarrow{\text{mux}} \boldsymbol{x}_{14} \xrightarrow{\text{ZOH}} \hat{\boldsymbol{x}} \xrightarrow{\text{LQR}} \boldsymbol{u} \xrightarrow{\tau_m} \boldsymbol{F} \quad \text{(closes loop)}$$

Each arrow represents a Drake system connection. There is no separate manipulator tracking loop — in this mode the manipulator *is* the cart actuator.